

HW10

1. A class is a blueprint or template that defines the data and behavior that an object will have. An instance of a class is an actual entity created from the blueprint that occupies space in memory.

3. Attributes are data stored in an object; methods are actions it performs.

4. Date ; default constructor

5. blueprint

9. When it is only intended to be called by other functions with in the same class

10. A constructor initializes an object upon creation a destructor clean up when it is destroyed.

21. class

22. Access specifiers

38. return

53. F

58. T

61. T

65. T

73. ① no () after class Triangle

② missing : after private and public

③ no return type of getsides()

④ no data type of s1, s2

⑤ two s2

⑥ missing ; after } of class

Programming challenges

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4  class Car
5  {
6  private:
7      int yearModel;
8      string make;
9      int speed;
10 public:
11     Car(int y, string m)
12     {
13         yearModel = y;
14         make = m;
15         speed = 0;
16     }
17     int getYearModel() const { return yearModel; }
18     string getMake() const { return make; }
19     int getSpeed() const { return speed; }
20     void accelerate()
21     {
22         speed += 5;
23     }
24
25     void brake()
26     {
27         if (speed >= 5)
28         {
29             speed -= 5;
30         }
31         else
32         {
33             speed = 0;
34         }
35     }
36 };
37 int main()
38 {
39     Car myCar(2024, "Porsche");
40
41     cout << "testing current speed..." << endl;
42     for (int i = 0; i < 5; i++)
43     {
44         myCar.accelerate();
45         cout << "current speed: " << myCar.getSpeed() << endl;
46     }
47
48     cout << "\ntesting brake function..." << endl;
49     for (int i = 0; i < 5; i++)
50     {
51         myCar.brake();
52         cout << "current speed: " << myCar.getSpeed() << endl;
53     }
54     return 0;
55 }
```

```
testing current speed...
current speed: 5
current speed: 10
current speed: 15
current speed: 20
current speed: 25
```

```
testing brake function...
current speed: 20
current speed: 15
current speed: 10
current speed: 5
current speed: 0
```

```
-----
Process exited after 0.2402 seconds with return value 0
请按任意键继续. . .
```